Science practice booklet: Grade 4 expectations

Grade 4 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 4 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: 4-LS1-1, 4-LS1-2

A worked example

A camel has wide, flat feet. Explain how this structure helps the camel survive in the desert.

1. Name the structure: wide, flat feet.
2. Think about the place where the camel lives: soft, loose sand.
3. Wide feet spread the camel's weight over a bigger area, so the feet do not sink deep into the sand.
4. Link it to survival: the camel can walk long distances to find food and water.

Wide, flat feet spread the weight so the camel does not sink in soft sand, and it can travel far to find food and water.

Questions

1. A camel's hump is not full of water. What does the hump store?

   A. Water

   B. Fat

   C. Sand

   D. Air

2. A camel has long eyelashes and can close its nostrils. What problem do these structures help with?

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3. The Arabian oryx has a pale, almost white coat. How does this help it in a hot, sunny desert?

   A. It reflects sunlight, so the oryx stays cooler

   B. It helps the oryx hide in green grass

   C. It keeps the oryx warm at night

   D. It lets the oryx see in the dark

4. Mangrove (القرم) trees grow in salty, muddy water along the coast, and their roots stick up out of the mud. Give one way this structure helps the tree.

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5. A falcon has a sharp, curved beak and strong claws. Which job do these structures do?

   A. Digging for water in the sand

   B. Swimming across the creek

   C. Catching prey and tearing food

   D. Keeping the falcon cool

6. What are two jobs of the roots of a date palm on Hind's family farm?

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7. Many desert animals come out to look for food at night. How does this help them survive?

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8. Match each sense organ to what it detects: eyes, ears, skin, tongue. Choose from: sound, light, taste, touch and heat.

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9. Hamdan sees a ball coming towards him and catches it. Put these steps in order: the brain decides what to do; the eyes detect the ball; the muscles move the arms; signals travel along nerves to the brain.

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10. Maryam touches a hot cup. Which body parts carry the signal from her skin to her brain?

    A. Bones

    B. Lungs

    C. Muscles

    D. Nerves

11. Salem says an animal's body parts help it survive best when they work together with its senses and brain. Do you agree? Explain with one example.

Section 2: Earth and space

Standards practised: 4-ESS1-1, 4-ESS2-1, 4-ESS2-2, 4-ESS3-1, 4-ESS3-2

A worked example

Over many years, heat in the day and cool at night make a rock crack into small pieces. Then the wind carries the small pieces away. Which part is weathering and which part is erosion?

1. Weathering is when rock is broken into smaller pieces.
2. Erosion is when the pieces are moved from one place to another by wind, water or ice.
3. Cracking into small pieces is weathering.
4. The wind carrying the pieces away is erosion.

Cracking is weathering; the wind carrying the pieces away is erosion.

Questions

1. A cliff has layers of rock that have not been disturbed. Which layer is the oldest?

   A. The top layer

   B. The middle layer

   C. The bottom layer

   D. All the layers are the same age

2. Rock layers in the Hajar mountains contain fossils of sea shells. What does this tell us about the place long ago?

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3. Wind blows sand from one side of a dune to the other side. What is this called?

   A. Weathering

   B. Erosion

   C. Melting

   D. Evaporation

4. Is each of these weathering or erosion? (a) Water freezes in a crack and splits a rock. (b) A stream carries pebbles down a mountain.

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5. The sand dunes near Hind's grandmother's house slowly change their shape and position over the years. What causes this?

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6. After heavy rain, water rushes down a dry valley (a wadi) and carries away soil and stones. Is this weathering or erosion? What may happen to the land over many floods?

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7. Ghaf trees and grasses grow on the edge of some dunes. How can their roots help the land?

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8. A map of the country has a key: blue is water, brown is mountains, yellow is sandy desert. Layla finds a large brown area on the map. What does it show?

   A. Water

   B. Sandy desert

   C. A city

   D. Mountains

9. Which of these is a renewable energy source?

   A. Sunlight

   B. Oil

   C. Natural gas

   D. Coal

10. A desalination plant uses a lot of energy to make drinking water from sea water. A solar park nearby could supply some of that energy. Give one good thing about using solar energy instead of burning oil or gas.

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11. A sandstorm is coming. Which action keeps people safest?

    A. Go outside to watch it

    B. Stay indoors and close the windows and doors

    C. Open the windows for fresh air

    D. Play on the dunes

12. Sand is blowing onto a road and covering it. Describe a solution that people could build to reduce the problem.

Section 3: Matter, forces and energy

Standards practised: 4-PS3-1, 4-PS3-2, 4-PS3-3, 4-PS3-4, 4-PS4-1, 4-PS4-2, 4-PS4-3, 3-5-ETS1-1, 3-5-ETS1-3

A worked example

A small solar-powered torch is left in the sun for the day. At night it gives light. Describe how energy changes form in the torch.

1. Start with the energy that comes in: light from the sun reaches the solar panel.
2. The panel changes the light energy into electrical energy, and this is stored in a battery.
3. At night the battery gives electrical energy to the bulb.
4. The bulb changes the electrical energy into light (and a little heat).

Sunlight (light energy) becomes electrical energy in the panel, is stored in the battery, and becomes light energy again in the bulb.

Questions

1. Omar rolls a ball slowly, then very fast, into a box of cardboard tubes. The fast ball pushes the tubes much further. What does this show?

   A. A fast ball has less energy

   B. A fast ball has more energy of motion

   C. A slow ball is heavier

   D. Speed does not change a ball's energy

2. When you rub your hands together quickly, they feel warm. What kind of energy changes into heat?

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3. A solar park is covered in solar panels. What energy change happens in a solar panel?

   A. Electrical energy into light energy

   B. Heat energy into sound energy

   C. Light energy into electrical energy

   D. Energy of motion into light energy

4. A spoon is left in a hot cup of karak. After a few minutes the handle feels warm. How did the heat get to the handle?

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5. A moving ball hits a ball that is resting. The resting ball rolls away. What happened to the energy?

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6. The wind turns a turbine in a wind farm. Which energy change happens?

   A. Light energy into electrical energy

   B. Electrical energy into energy of motion

   C. Heat energy into light energy

   D. Energy of motion into electrical energy

7. A floating ball bobs up and down on sea waves. Does the wave carry the ball all the way to the beach? Explain.

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8. In a lit room, Khalid can see a book on the table. Why can he see it?

   A. His eyes send out light to the book

   B. The book makes its own light

   C. Light bounces off the book into his eyes

   D. The dark helps light reach the book

9. Two cousins plan to send messages across the garden with a torch: two flashes mean yes and three flashes mean no. Why must they agree on the pattern first?

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10. Name one other way, besides flashing a light, that a pattern of sound or light can be used to send information.

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11. Noura is designing a shade for the school playground. It must keep the sun off the children, and it must cost less than AED 500. Which one is a constraint (a limit)?

    A. It must cost less than AED 500

    B. It must keep the sun off the children

    C. It must be a shade

    D. It must have a strong frame

12. Khalid tests two shade designs by measuring the temperature under each one. Why should he test both in the same place and at the same time of day?

Answer guide for parents

Section 1: Living things

1. B. Fat. The hump stores fat that the camel uses for energy when food is scarce. If your child chose A (Water), they believe the common myth, but the hump holds fat.
2. Long eyelashes shield the eyes and closed nostrils stop sand getting in. Both stop blowing sand from hurting the camel in a sandstorm.
3. A. It reflects sunlight, so the oryx stays cooler. A pale coat reflects sunlight, so the oryx heats up less. If your child chose B (hide in green grass), they forgot the desert has little green grass.
4. Roots hold the tree steady in soft mud, and they let it take in air because the mud holds little. Either one is a good answer.
5. C. Catching prey and tearing food. Strong claws grab the prey and the sharp beak tears the food. If your child chose A (digging for water), they matched the claws to digging, but a falcon hunts.
6. Roots take in water from the soil and hold the tree firmly in place. The water goes up to the leaves and fruit, and the grip keeps the palm from falling.
7. The night is cooler, so animals lose less water from their bodies. They avoid the strong heat of the day and still find food.
8. Eyes detect light, ears detect sound, skin detects touch and heat, and the tongue detects taste. Each organ matches one kind of information from the world.
9. First the eyes see the ball, then nerves carry the signal to the brain. The brain decides, then the muscles move the arms.
10. D. Nerves. carry signals between the skin and the brain. If your child chose C (Muscles), they mixed up carrying signals with moving the body.
11. Yes. A falcon's eyes spot prey, the brain decides, and wings and claws catch it. The parts only work well because they work together.

Section 2: Earth and space

1. C. The bottom layer. Layers form from the bottom up, so the bottom layer formed first. If your child chose A (The top layer), they picked the newest layer.
2. Sea shell fossils mean the land was once under the sea. Shells only get buried in rock where there was sea water long ago.
3. B. Erosion. Wind moving sand from one place to another is erosion. If your child chose A (Weathering), they mixed up breaking rock with moving pieces.
4. (a) Weathering, because the ice splits the rock into pieces. (b) Erosion, because the stream moves the pebbles to a new place.
5. Wind lifts sand grains, carries them and drops them somewhere new. Over many years this moves the dune and changes its shape.
6. Erosion, because moving water carries soil and stones away. After many floods the wadi gets deeper and wider as more land is carried off.
7. Roots hold the soil and sand in place. Less is carried off by wind and water, so the land wears away more slowly.

8. D. Mountains. The key says brown means mountains, so a big brown area is mountains. If your child chose B (Sandy desert), they mixed up brown with yellow.
9. A. Sunlight. will not run out, but oil, gas and coal can. If your child chose B (Oil), they picked a fuel that people use a lot, but it is not renewable.
10. Sunlight does not run out, and solar panels give off no smoke. Burning oil or gas uses up fuel and makes gases. One good thing is enough.
11. B. Stay indoors and close the windows and doors. Closed windows and doors keep the blowing sand out. If your child chose A (Go outside to watch it), they were curious, but sand can hurt eyes and lungs.
12. A fence or a row of plants beside the road blocks the wind. The sand drops there, so less of it reaches the road. Any sensible idea works.

Section 3: Matter, forces and energy

1. B. A fast ball has more energy of motion. A faster ball has more energy of motion, so it pushes the tubes further. If your child chose A (less energy), they thought speed takes energy away.
2. Energy of motion changes into heat. The faster you rub, the more heat the movement makes, so your hands warm up.
3. C. Light energy into electrical energy. A solar panel changes light energy from the sun into electrical energy. If your child chose A (Electrical into light), they reversed it, which is what a lamp does.
4. Heat moves along the metal spoon from the hot karak to the cooler handle. Heat always moves from hot to cooler.
5. Some energy moved from the first ball to the second ball. That is why the second ball began to move.
6. D. Energy of motion into electrical energy. The wind has energy of motion, and the turbine changes it into electrical energy. If your child chose B, they reversed it, which is what a motor does.
7. No. The water moves up and down as the wave passes, so the ball stays near the same place. Only the wave's energy travels to the beach.
8. C. Light bounces off the book into his eyes. We see a book when light bounces off it into our eyes. If your child chose A (his eyes send out light), they think seeing works like a torch, but eyes only receive light.
9. They must agree first so each pattern has a meaning. Without that, two flashes could mean anything, so no message is passed.
10. A car horn pattern, a siren or a school bell pattern all send information. The sound or light pattern tells people what to do. Any sensible answer works.
11. A. It must cost less than AED 500. A constraint is a limit, and costing less than AED 500 is a limit. If your child chose B (keep the sun off), that is the criterion, what the design must do.
12. The test is fair when only the design changes. Same place and same time mean any temperature difference comes from the design.